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Palaeomagnetic constraints upon the palaeogeographic position of the Baltic Shield in the Ordovician

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ABSTRACT

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The palaeomagnetic analysis of the Ordovician limestones in Southern Scandinavia (75 sites) has given the following results: (1) in the southern part of the Oslo Region, all Ordovician sediments were remagnetized during the Lower Permian, by a thermo-chemical mechanism triggered by the emplacement of the Permian volcanic and intrusive suite; (2) in the northern part of the Oslo Region, the Upper Caradoc Mjøsa Limestone carries a high-temperature component (decl. = 53°, incl. = −40°, $\alpha_{95} = 25°$, 4 sites) with corresponding palaeopole at 4° S, 143° E; (3) in southern Sweden, all Lower Ordovician "*Orthoceras* limestones" sampled in Dalarna, Närke, Scania, Väster-götland, Öland and Öster-götland display the same well-defined southeast steep high-temperature components, leading to a palaeopole at 30° N, 55° E, $A_{95} = 9°$; in areas with late Palaeozoic volcanic activity (Scania, Väster-götland), this high-temperature component is partially overprinted by a remagnetization associated with dolerite emplacement. According to these results, Baltica was situated in temperate southern latitudes (50° S) in the Early Ordovician and has drifted northwards at a latitudinal drift rate of 4 to 7 cm/year to reach tropical latitudes (20° S) by the Late Ordovician to Late Silurian. This interpretation is in agreement with previous models based on the occurrence of Ordovician reef structures, and sheds new light on the early Palaeozoic position of Baltica relative to other continental blocks.

Introduction

The reconstruction of the Palaeozoic geodynamic evolution which ended with the constitution of Pangea in the Permian, entails precise knowledge of the successive latitudinal positions of the major individual continental plates. This can be derived from palaeomagnetic, palaeobiogeographic and palaeoclimatic evidence and a large number of reconstructions have already been published during the past two decades. Despite the continued generation of new data that have led to successive modifications of these reconstructions, a general agreement about the Palaeozoic evolution has yet to be reached (Scotese and McKerrow, 1990). One of many problems and still an open question, is the knowledge of the precise position of Baltica (Baltoscandia and East-European–Russian platform) during the early Palaeozoic. This has never been precisely constrained, hence new data are of great interest (Torsvik et al., 1990, p. 38).

The aim of the present paper is to present and discuss palaeomagnetic results obtained from Ordovician limestones in Sweden and Norway and to show how these may help solve the problem of the position of Baltica. To begin with we present a short review of the present state of knowledge

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based on palaeobiogeographical evidence and hitherto preliminary and incomplete palaeomagnetic data.

Palaeobiogeography

Spjeldnæs (1961) suggested that Ordovician faunal provinces correspond to distinct climatic zones, and later papers developed this idea to produce palaeogeographical reconstructions based on evidence from the study of sediments and benthic faunas. In these, the Baltic province has always been considered different from the Gondwana and North American provinces. This has been particularly well illustrated for trilobites (Whittington and Hughes, 1972) and brachiopods (Williams, 1973; Jaanusson, 1973; Neuman, 1984; Bruton and Harper, 1985). The isolation of these provinces is thought to be related to climatic zonation and to oceanic barriers, the Proto-Atlantic (Iapetus) Ocean and the mid-European Ocean separating them. Subsequent studies and more detailed analysis of the Ordovician sedimentary successions and faunas did not upset these reconstructions which have only been slightly modified, more fully illustrated, and more precisely delimited. At present there is a general consensus about the global palaeogeographic configuration for the Ordovician Period despite some discrepancies as those concerning the existence of one or two successive mid-European oceans (see for example Cocks and Fortey, 1982; Robardet et al., 1990; Paris and Robardet, 1992). However, the Ordovician latitudinal position of Baltica cannot be defined precisely by faunal and sedimentary evidence. Some reconstructions have shown the Baltic continent at equatorial latitudes (Whittington and Hughes, 1972) during the entire Ordovician, whereas other authors are rather inclined to propose a more temperate position (Spjeldnæs, 1961). More precise discussions were based upon the sedimentological characteristics of the platform carbonate deposits which developed in the Baltic regions throughout the Ordovician. Detailed studies of the components of the Ordovician limestones, and particularly the existence or the absence of carbonate grains, pellets and ooids, suggest that these sediments were deposited: (1) in a temperate climatic zone during Early and early Middle Ordovician times; (2) in a subtropical or tropical zone during the late Middle and Late Ordovician (Jaanusson, 1972; Lindström, 1984).

These conclusions are corroborated by the occurrence of organisms dependent of warm water and low latitudes, such as stromatoporoids (Webby, 1980), and by the existence of reefs and associated bahamitic sediments in the upper part of the Caradoc and in the Ashgill Series. The temporal distribution of the carbonate types and of the coral–stromatoporoid reefs indicates that the North-European Baltic plate has probably drifted northwards from temperate to middle–low latitudes during the Ordovician (Webby, 1984).

Palaeomagnetism

As stressed by Bylund (1986), palaeomagnetic data concerning the Baltic regions during the early Palaeozoic are rare and not fully conclusive. Preliminary results published only as short abstracts have indicated that the Baltic Shield was located at about 60 ° S during the Early and early Middle Ordovician and then moved rapidly to a near equatorial position (Noltimier and Bergström, 1976; Bergström and Noltimier, 1982), but these conclusions have never been completely documented and discussed.

A thorough palaeomagnetic investigation of the Ordovician limestones in Sweden (mainly the Early Ordovician "*Orthoceras* limestone") (Claesson, 1977), revealed a steeply inclined bulk direction of magnetization, but the interpretation of these results was not definitely established as this magnetization had a multivectorial character and proper demagnetization techniques were not available to decipher it. A reinterpretation of Claesson's pilot samples was recently published (Torsvik and Trench, 1990) and a new pole position for the Swedish "*Orthoceras* limestone" was identified, using only Claesson's samples which were properly demagnetized. This new pole is only slightly offset from the initial Claesson's pole, confirming the rather high (~ 50 ° S) palaeolatitude of southern Sweden in the Early Ordovician. However, it should be noticed that this

reinterpretation is based upon Claesson's laboratory work of the 1970's, and has therefore not benefited from improved laboratory techniques developed in the 1980's, with the introduction of cryogenic magnetometers.

A review of palaeomagnetic data from the Soviet Union (Khramov et al., 1981) included some information concerning the East-European–Russian platform which represents the eastern extension of the Baltic regions. The "Russian plate" was located in the Southern Hemisphere at low (20 to 30°) latitudes during the Late Cambrian and the Early Ordovician. However, these data might reflect late Palaeozoic overprinting which is frequently observed in other European regions (Torsvik et al., 1990). Most frequently, the position of Baltica is based on palaeomagnetic data from England or Wales, areas peripheral to Baltica. Thus in all the recently published reviews concerning the Caledonide Orogen and related areas (Piper, 1985; Briden et al., 1988; Torsvik et al., 1990), the latitudinal position of Baltica is not clearly fixed until at least the Late Silurian.

Geological setting and sampling

The relatively thin lower Palaeozoic succession of Baltoscandia (Fig. 1) consists mainly of limestones and mudstones, now shales, deposited on a stable platform that includes the "Russian platform". The sediments are arranged in distinct belts known as confacies, each with a distinct fauna and lithology (Jaanusson, 1976).

In Sweden, the studied areas belong to the autochthon of the Scandinavian Caledonides (Bruton et al., 1985) in which sediments, about 200 m thick, are not folded in the south and lie flat or dip gently. Today they are preserved in small outliers often fault-bounded and capped by Permian sills (Fig. 1). In the Siljan district to the north, lower Palaeozoic rocks dip steeply and form a complex ring structure, interpreted as marginal to an impact crater (see Jaanusson, 1982b for references).

In the Oslo Region, Norway (Fig. 2), within the Oslo graben, the Cambro–Silurian succession is about 2000 m thick. In this area, located near the Caledonian front, the lower Palaeozoic suc-

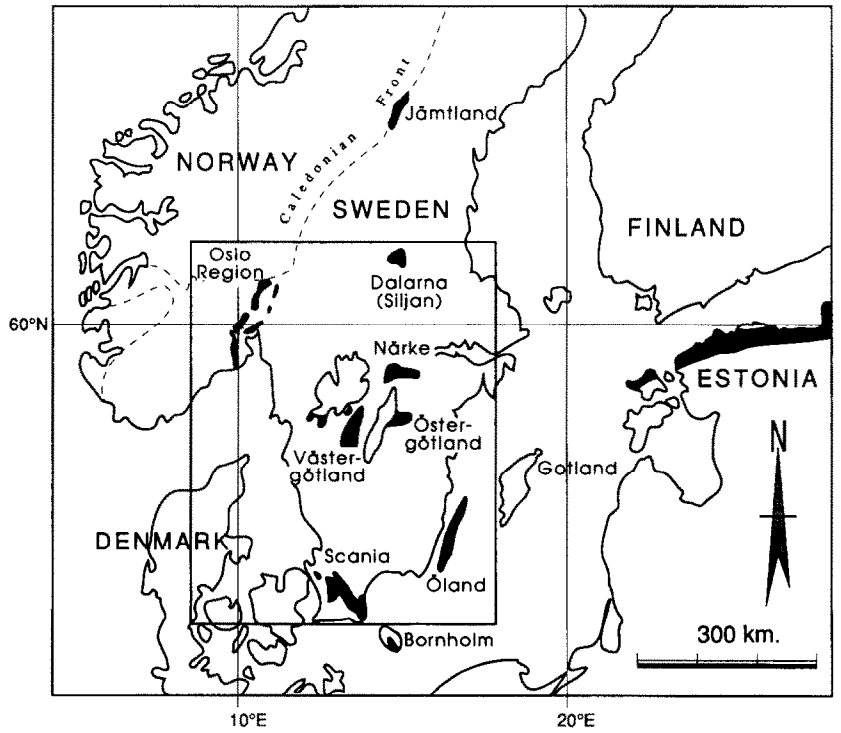


Fig. 1. Map of the Lower Palaeozoic rock occurrences in southern Scandinavia. The rectangle delimits the area of this study.

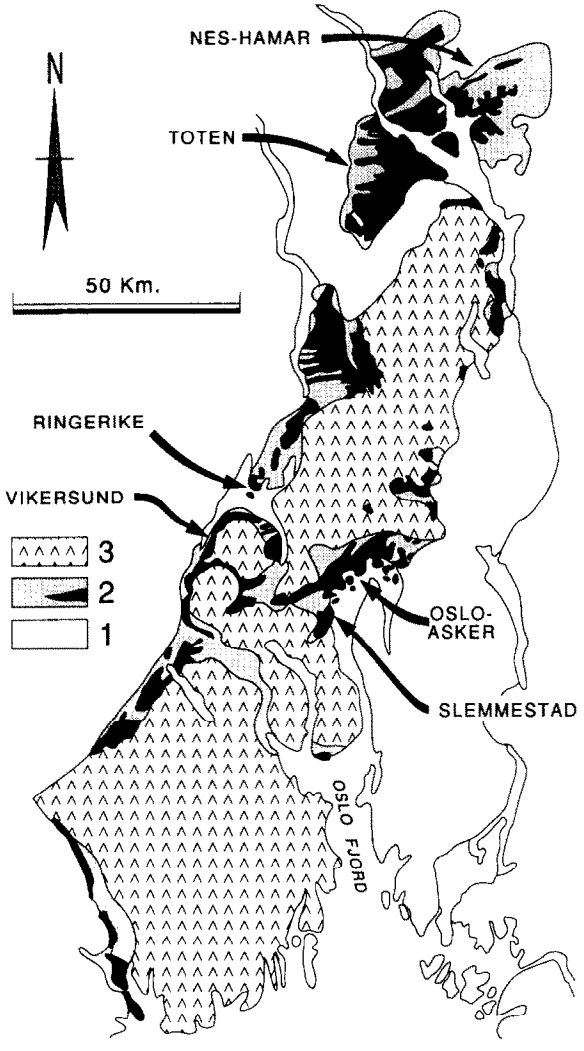


Fig. 2. Simplified geological map of the Oslo Region, with the localisation of the sampling areas: *1* = Proterozoic basement; *2* = Lower Palaeozoic sediments, including the Ordovician in black; *3* = Permian igneous bodies.

cession, up to and including the Ringerike Sandstone Formation of Wenlock–Ludlow age, has been folded and thrust-faulted above a decollement at the top of the Cambrian Alum Shale Formation. It is therefore regarded as parautochthonous (Bockelie and Nystuen, 1985). The Permian igneous activity was particularly important in this area.

The autochthonous Ordovician in Norway and Sweden

During the Ordovician, an epicontinental sea extended over the entire Baltic platform from west of the Oslo Region to the Moscow basin in the east (Bruton and Williams, 1982; Bruton, 1989). Most of the areas sampled are within the "Central Baltoscandian belt" of Jaanusson (1976, 1982a), where carbonate sediments predominate, and the "Scanian confacies belt" characterized by terrigenous sediments. Sequences are thin and the total thickness generally does not exceed 150–200 m in Sweden, although this figure is more than doubled in some parts of the Oslo Region, Norway (Bruton and Owen, 1982; Bruton et al., 1985). The succession includes several breaks caused by non-deposition and erosion, but the most striking feature is the extremely low average rate of deposition of 1–3 mm per 1000 years (Jaanusson, 1982a).

In the Oslo area, almost all the stratigraphic units were sampled (cf. Appendix), whilst in the Mjøsa area, sampling was mainly from the Mjøsa Limestone of Caradoc age. In Sweden, sampling was restricted to the Lower Ordovician "*Orthoceras* limestones" of Arenig–Llanvirn age. Here, sampling sites were chosen with reference to those used in the study by Claesson (1977) in order to allow a comparison of results.

Conodont colour indices (CAI) were also used to determine sampling sites since these provide a useful guide to palaeotemperatures and the thermal effects resulting from sedimentary burial and igneous activity (see Armstrong and Strens, 1987; Nowlan and Barnes, 1987). This method has demonstrated that, in the Baltic regions, Permian igneous activity has been the main cause of thermal effects in the Ordovician and Silurian rocks (Bergström, 1980; Aldridge, 1984, 1986). These studies supply precise CAI values for the main Ordovician areas, with high values (4.5–5) both in the Oslo–Ringerike region (Norway) and in Scania (south Sweden). Very low values (1–1.5) have been reported from Öland, Öster-götland, Närke and Dalarna. In Väster-götland, both low and high (up to 6–7) values are known, depending of the proximity of Permian igneous rocks. Results from the Mjøsa district are not uniform, some Silurian rocks (Aldridge, 1984) showing lower values than the Ordovician (Bergström, 1980). Nevertheless, these data give important clues for determination of the areas where one can expect

that there was no major thermal alteration of the original magnetization of the rocks. The sampling in areas with various CAI values was intended to relate the development of Permian magnetic overprinting to the estimated palaeotemperatures.

Palaeomagnetic analysis

Magnetic measurements were done using both the Schonstedt DSM-1 spinner magnetometer or a LETI cryogenic magnetometer. Thermal demagnetization was conducted with a Schonstedt TSD-1 oven, and alternating fields (af) demagnetization with Schonstedt GSD-1 equipment. Residual fields in magnetic shields were always less than 10 nT. All treated samples were progressively demagnetized, and low field susceptibility was checked after each temperature step, using the Bartington MS-2 instrument. Demagnetization trajectories were interpreted using a computer-assisted graphical system (Perroud, 1985), coupled with the mathematical principal component analysis (Kirschvink, 1980). On the basis of these studies, three areas with consistent behaviour were identified: Oslo Region south, Oslo Region north, and south Sweden.

Oslo Region south

This area includes sites localised in Oslo–Asker, at Slemmestad, near Vikersund and in Ringerike (Fig. 2). Sites with various dips were sampled as was a metric fold and clasts from a

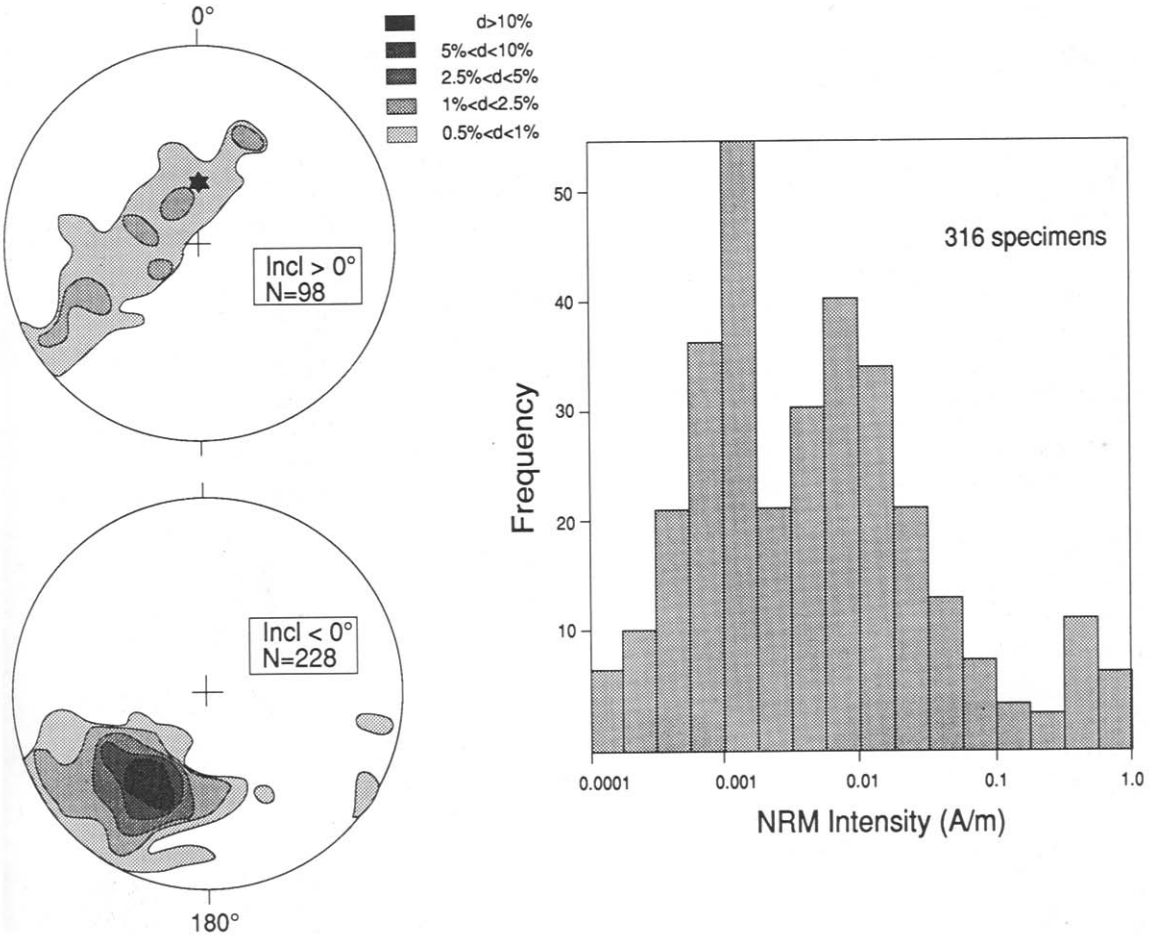


Fig. 3. NRM results from the Oslo Region south. Left: density contour plot of the NRM directions (d is the density of observations per unit area). Right: frequency distribution of NRM intensities.

TABLE 1

Palaeomagnetic results from Oslo Region south

Sites	$Decl._{IS}$	$Incl._{IS}$	n	k	α_{95}	$Decl._{TC}$	$Incl._{TC}$	Observations
Oslo–Asker								
OA	198	−41	3	231	8	179	−10	
OB	201	−48	3	331	7	–	–	dyke
OC	201	−41	3	374	6	181	−11	
OD	215	−53	3	54	17	178	−26	OB contact
OE	199	−39	3	347	7	182	−10	
OF	207	−38	2	–	–	186	−9	
OG	207	−47	4	249	6	196	−53	
OH	209	−50	3	228	8	191	−43	
OI	203	−48	2	–	–	192	−25	
OJ	210	−42	3	397	6	308	−62	
OK	212	−49	3	472	6	190	−32	
OL	216	−50	2	–	–	192	−33	
OM	215	−49	3	1631	3	194	−36	
ON	198	−43	1	–	–	186	+22	
OO	198	−32	3	397	6	198	+31	
OP	206	−48	3	704	5	186	+17	
OQ	215	−47	5	400	4	190	+14	Conglomerate
Mean IS	206	−45	17	127	3			
Mean TC			16	6	16	189	−18	excluding dyke
Slemmestad								
OR	204	−39	1	–	–	177	−22	
OS	–	–	–	–	–	–	–	unstable
OT	218	−49	2	–	–	298	−78	
OU	216	−44	4	113	9	202	−41	
OV	213	−38	3	462	6	198	−26	
OW	213	−48	3	222	8	197	−34	
OX	220	−36	3	375	6	201	−14	
OY	211	−44	3	514	5	192	−15	
OZ	213	−48	3	525	5	191	−16	
O1	210	−39	3	210	8	193	−20	
O2	209	−49	3	412	6	186	−29	
O3	210	−47	3	500	6	187	−29	
O4	213	−51	4	1144	3	187	−28	
Mean IS	213	−44	12	183	3			
Mean TC			12	14	12	193	−30	
Ringerike								
RA	20	+44	1	–	–	–	–	dyke
RA	22	+36	5	921	3	17	+32	contact
RB	197	−46	4	178	7	219	−58	fold($k_{TC} = 7$)
RC	191	−44	3	232	8	186	−37	
RD	185	−50	4	637	4	214	−56	
RE	189	−45	4	218	6	220	−53	
RF	185	−44	4	95	10	198	−46	
RG	188	−43	4	417	5	203	−43	
Mean IS	192	−44	8	170	4			uniform polarity
Mean TC			7	39	10	204	−47	excluding dyke

TABLE 1 (continued)

Sites	$Decl._{IS}$	$Incl._{IS}$	n	k	α_{95}	$Decl._{TC}$	$Incl._{TC}$	Observations
Vikersund								
RH	194	−32	2	–	–	206	−28	
RI	202	−49	4	327	5	187	−51	
RJ	198	45	5	196	6	335	−28	
RK	198	−50	4	175	7	195	−23	
RL	–	–	–	–	–	–	–	unstable
Mean IS	198	−44	4	88	10			
Mean TC			4	3	74	216	−49	

Notes: Decl. = declination; Incl. = inclination; IS = in situ; TC = tilt corrected; n = number of samples; k, α_{95} = Fisher statistical parameters; see Appendix for site locations.

channel-fill conglomerate of Ashgill age in order to use field tests. We have also sampled a few Permian doleritic dikes and their contact to check for the influence of the Permian volcanism.

The intensity of initial magnetization varies considerably between sites, from 10^{-4} A/m up to 1 A/m, which is anomalously high for limestone samples. Furthermore, NRM directions show a very strong grouping around a south-southwest and upward mean direction, together with a clear planar distribution between that direction and the present-day field direction (Fig. 3). According to Bylund (1986), the Early Permian field direction in the Oslo basin (lat. 60° N, long. 10° E) was decl. = 204°, incl. = −40° (reverse polarity). The distribution of the NRM directions appears therefore to be closely controlled by both the Permian and the present-day fields.

Demagnetization experiments have totally confirmed this interpretation. In nearly all samples, the dominant magnetization component has the same SSE and upward direction, irrespective of rock type or tilt. It is generally isolated between 200° C and 500° C, and converges towards the origin in orthogonal projections (Fig. 4). Due to this systematic behaviour, we have treated only 3 to 5 samples in each site, even when much more were sampled. The site-mean estimations of the characteristic magnetizations are listed in Table 1. In some sites, a soft component is eliminated below 200° C, the direction of which being similar to the present-day field direction. In the whole collection there is no indication of a possible component of magnetization with high (> 500° C) unblocking temperatures.

Apart from this general behaviour, a few interesting features deserve to be noted:

(1) Two sites in the Ringerike area display a NNE and down direction of magnetization (Fig. 4, RA235), exactly antiparallel to the usual SSE and up directions. It concerns one Permian dyke and limestones at its contact.

(2) Unblocking temperatures near 300–350° C, indicating the presence of pyrrhotite, are observed (Fig. 4, OX184), sometimes together with the usual 500–550° C unblocking temperatures. This is specially true in the Slemmestad area, and is accompanied by a very strong hardness to af demagnetization (Fig. 4, OZ194). For this reason, af demagnetization was not very successful in isolating components of magnetization. On this basis, systematic thermal demagnetizations were conducted and the results given in Table 1 rely only on the thermal experiments.

All the field tests we have conducted, lead to the conclusion that the observed characteristic magnetization is a Permian remagnetization:

(1) The conglomerate test of site OQ (Table 1) is negative. All pebbles reveal the same direction of magnetization

(2) All fold tests are negative. In every area, significative decreases of k are observed upon tilt correction (Table 1). In site RB, a metric fold was sampled, which reveals again a negative fold test (k decreases from 178 to 7, $n = 4$).

(3) When Permian dykes were sampled, we

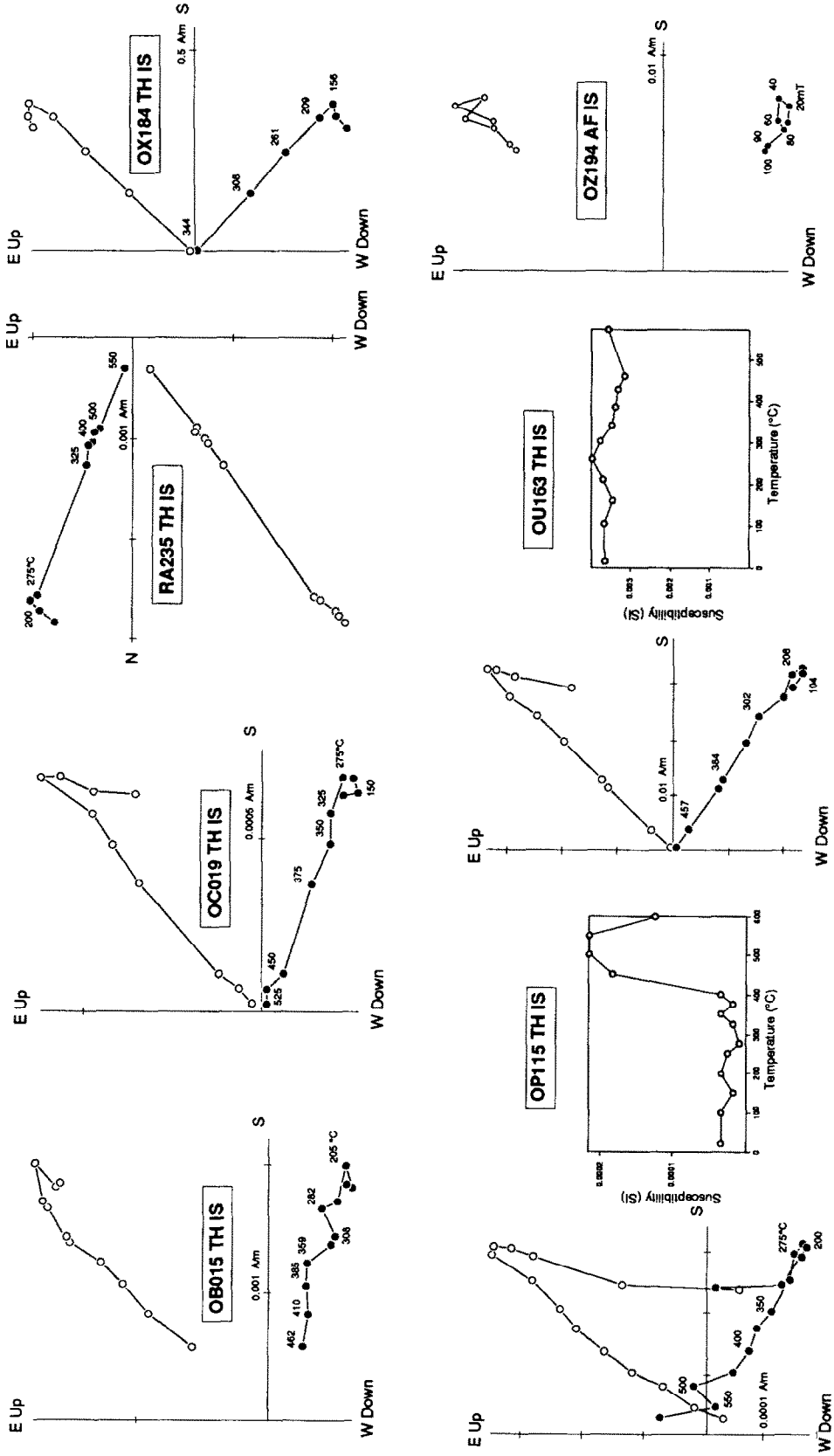


Fig. 4. Orthogonal projections of demagnetization curves from the Oslo Region south. Dots (circles) correspond to projections in the horizontal (vertical) plane; *TH*, *AF* = thermal and alternating field demagnetization, respectively; *IS* stands for in situ; for site location see Appendix.

observed the same magnetization in the dyke, in the contact, and in the unheated host rock (see for example dyke sample OB015 and limestone OC019, Fig. 4).

(4) Finally, the observed directions are quite consistent with the expected Early Permian direction of magnetization in the Oslo Region. The overwhelming majority of reverse magnetization suggests that acquisition took place in the Kiaman Magnetic Interval.

The origin of this magnetization is clearly related to the Permian volcanic activity in the Oslo Region. However, the mechanism for its acquisition can be either purely thermal (that is without any change in the rock mineralogy), purely chemical (without any increase of temperature) or thermo-chemical (mineralogical changes triggered by temperature increase). The CAI indexes clearly reveal that temperatures up to 300 °C were reached in most Ordovician rocks of the area, so the purely chemical mechanism can be excluded. However, there is some evidence that mineralogical changes took place in the rocks as exemplified by the demagnetization diagrams of Fig. 4. Two types of behaviour are observed: sample OU163 NRM is largely dominated by the Permian overprint; it reveals only a very small soft viscous component, and its susceptibility does not vary in the course of the thermal demagnetization experiment. In contrast, sample OP115 has a much lower intensity, a much stronger soft component and displays large changes in susceptibility after 400 °C, which perturbs significantly the demagnetization trajectory at high temperatures. It appears therefore that the remagnetization process, when it is well developed, stabilizes the magnetic mineralogy with respect to temperature.

In the Oslo–Asker area, the intensity of remagnetization varies between the sites ("remagnetization gradient"). To estimate the degree of remagnetization, we can use several methods: (a) one can use the angular deviation (delta) between the NRM direction and the Permian expected direction; the greater the degree of remagnetization, the less that angle; (b) another approach consists in using the Fisher dispersion parameter k of the NRM site-mean directions; the higher

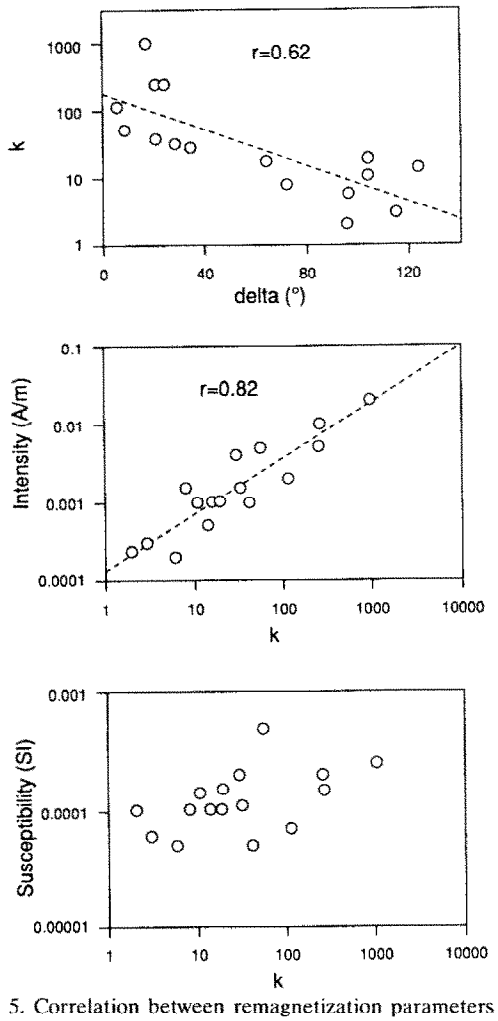


Fig. 5. Correlation between remagnetization parameters (see text) from the Oslo–Asker area.

the degree of remagnetization, the higher the k value. As shown in Fig. 5, the two parameters k and delta are well correlated.

A comparison of intensity of magnetization, initial susceptibility and degree of remagnetization (Fig. 5) reveals that the intensity of NRM is correlated with the k parameter, while the susceptibility is not. The same observation can be made using the delta angle introduced above. It seems therefore that the remagnetization process is not due to an enrichment in the iron content of the limestones, what should produce an increase of susceptibility, but rather to a re-distribution of iron within the mineralogical assemblage, which produces ferromagnetic phases (magnetite, pyrrhotite etc.) from paramagnetic ones (micas,

pyrite etc.). Such a mechanism does not require fluid percolation. It corresponds rather to the chemical response of the geological material to the thermal pulse produced by the Permian volcanism.

Oslo Region north

The Upper Caradoc Mjøsa Limestone was sampled in two large quarries in the Toten and Hamar areas (5 sites), together with the Arenig–Llanvirn "*Orthoceras* limestone" (1 site). NRM measurements are very different when compared with those from the Oslo Region south (Fig. 6): directions appear to be much more influenced by the present-day field than by the Permian overprint; intensities are lower (10^{-3} A/m) and a short-period viscosity test reveals that the viscous magnetization is quite high (around 60 to 80%). According to previous results, it seems than the Permian overprinting mechanism was much less efficient in this area. We should therefore expect that older components of magnetization have been preserved in these samples. This was confirmed by the demagnetization experiments described below.

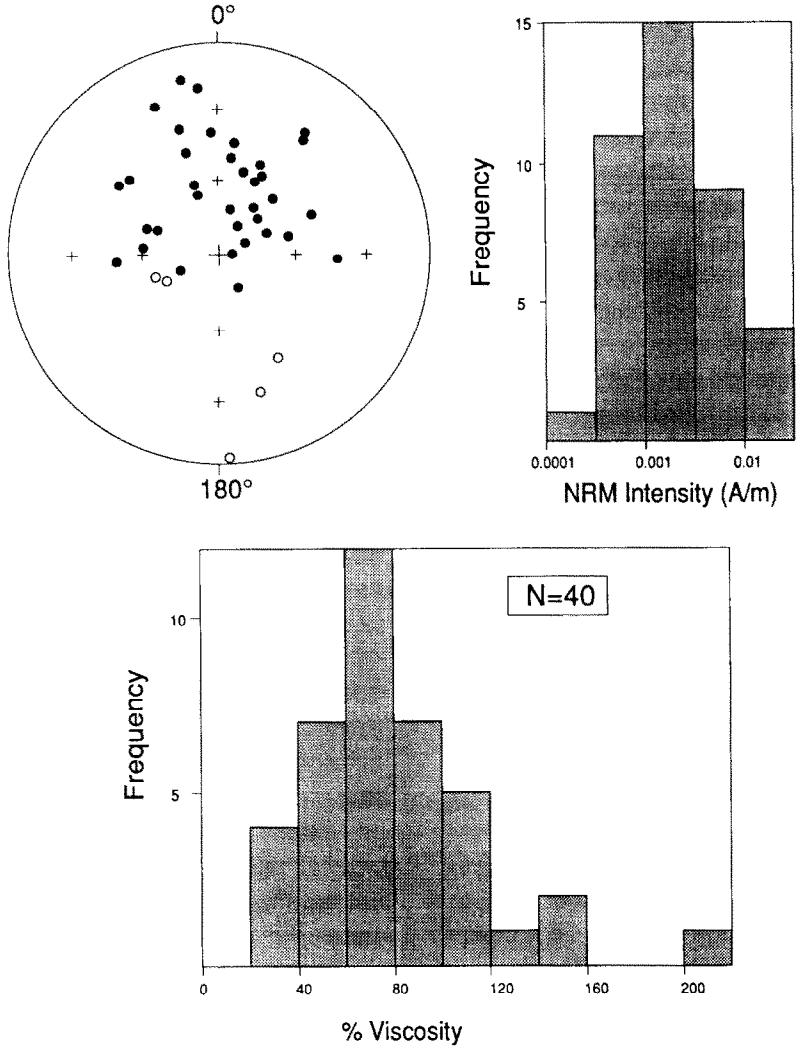


Fig. 6. NRM results from the Oslo Region north. Top left: stereographic projection of NRM directions (dots: lower hemisphere; circles: upper hemisphere). Top right: frequency distribution of NRM intensities. Bottom: frequency distribution of viscosity indices.

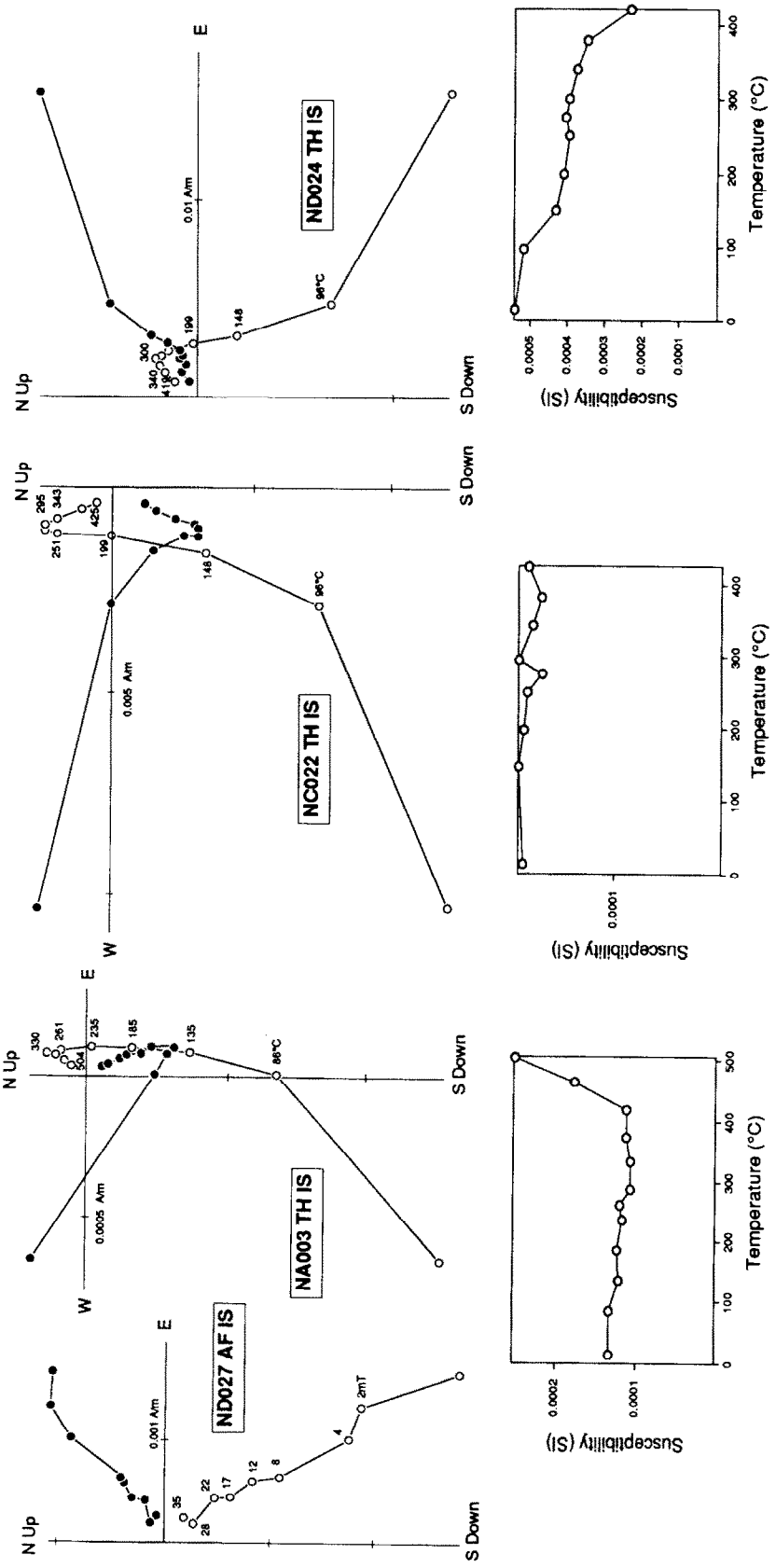


Fig. 7. Orthogonal projections of demagnetization curves from the Oslo Region north. Same legend as Fig. 4.

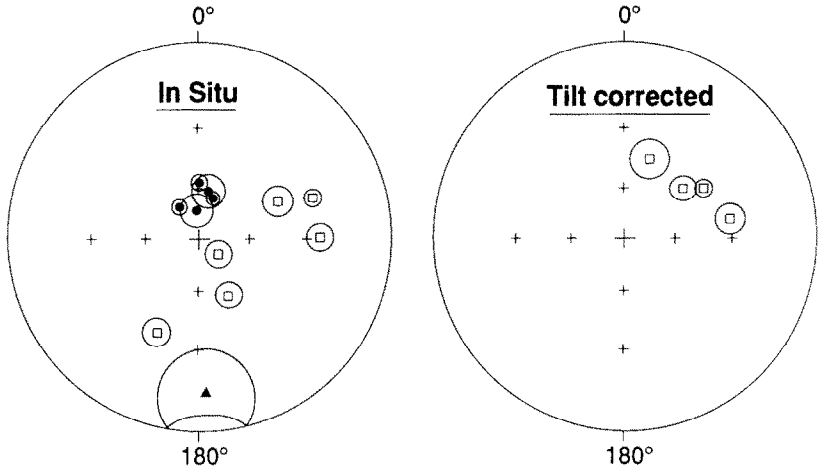


Fig. 8. Stereographic projections of the palaeomagnetic site-mean directions, with confidence circles, from the Oslo Region north, before (left) and after (right, high temperature components only) tilt correction.

Thermal progressive demagnetizations show the multivectorial nature of the NRMs (Fig. 7): apart from viscous components eliminated by 100 °C, most samples reveal a soft component, steeply inclined and pointing north, eliminated between 100 and 300 °C, and which represents the larger part of the NRM. High-temperature components can be recovered after its elimination, between 300 and 500 °C. In site NC, this last component is parallel to the Permian field

TABLE 2

Palaeomagnetic results from the Oslo Region north

Sites	$Decl._{IS}$	$Incl._{IS}$	n	k	α_{95}	$Decl._{TC}$	$Incl._{TC}$	Observations
Mjøsa limestone								
NA	356	+74	9	32	9	202	+38	100–250 °C
NA	178	+13	9	9	18	198	−50	250–400 °C
NA	153	−54	7	84	7	50	−46	400–500 °C
NB	128	−75	8	52	8	18	−39	400–550 °C
NC	327	+69	7	255	4	195	+32	100–300 °C
NC	205	−34	7	90	6	317	−61	300–400 °C
ND	10	+62	7	42	9	244	+68	100–350 °C
ND	65	−41	5	108	7	81	−30	350–450 °C
NE	18	+65	10	176	4	73	+63	100–350 °C
NE	71	−26	8	211	4	57	−36	350–450 °C
"Orthoceras limestone"								
NF	0	+57	10	125	4	357	25	100–350 °C
NF	90	−25	8	97	6	102	−17	350–400 °C
Means								
Low T ° IS	0	+66	5	68	9			NA-NC-ND-
Low T ° TC			5	3	59	215	+76	NE-NF
High T ° IS	92	−55	4	6	41			NA-NB-ND-
High T ° TC			4	14	25	53	−40	NE

Notes: see Table 1 for explanations of abbreviations.

(Fig. 7, NC022), while in all other sites, it is very different, in the northeast or southeast quadrant, with moderate to high upward inclinations (Fig. 7, NA003, ND024). In site NA, an intermediate component can be determined in a narrow temperature interval, with a south and shallow direction (Fig. 7, NA003). However, it presents a large scatter and a direction which is intermediate between the low- and high-temperatures components (Fig. 8) and may therefore correspond to their composition. Af demagnetization (Fig. 7, ND027) was inefficient in separating the components, and thus only thermal results were used to obtain the site-mean results listed in Table 2. It can be noted that the susceptibilities display some changes in the course of the thermal demagnetization (Fig. 7), except in the case of site NC.

As shown in Fig. 8, the soft components are very well grouped in field coordinates (in situ). The mean direction is decl. = 0°, incl. = +66°, $\alpha_{95} = 9°$ (5 sites). When correcting for tilt, the site-mean directions diverge (k decreases from 68 to 3), indicating a post-folding origin. According to its direction, this component very probably represents a present-day field remagnetization. The site NC high-temperature component is very similar to all Permian remagnetizations observed in the south of the Oslo Region, and was not included in the evaluation of the mean. High-temperature components from the other sites of

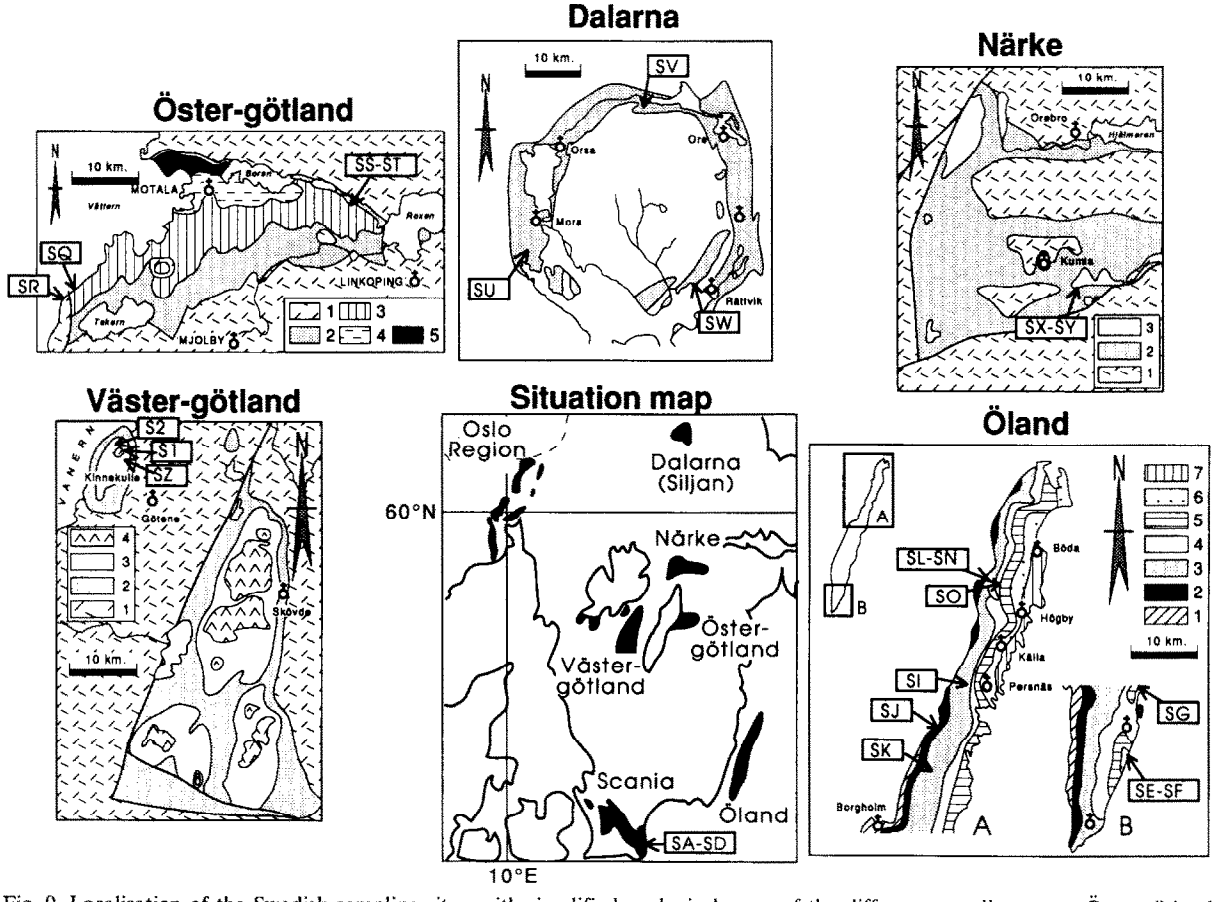


Fig. 9. Localisation of the Swedish sampling sites, with simplified geological maps of the different sampling areas. Öster-götland: *1* = Proterozoic; *2* = Cambrian–Tremadoc; *3* = "*Orthoceras* limestone"; *4* = Middle Ordovician limestones and shales; *5* = Silurian. Väster-götland: *1* = Proterozoic; *2* = Cambrian; *3* = Ordovician–Silurian; *4* = Permian igneous rocks. Närke: *1* = Proterozoic; *2* = Cambrian; *3* = Lower Ordovician limestones. Öland: *1* = Alum shale; *2* = "*Planilimbata* limestone"; *3* = "*Asaphus* limestone"; *4* = "*Gigas* limestone"; *5* = Skarlöv Limestone; *6* = Källa Limestone; *7* = "*Chasmops* limestone".

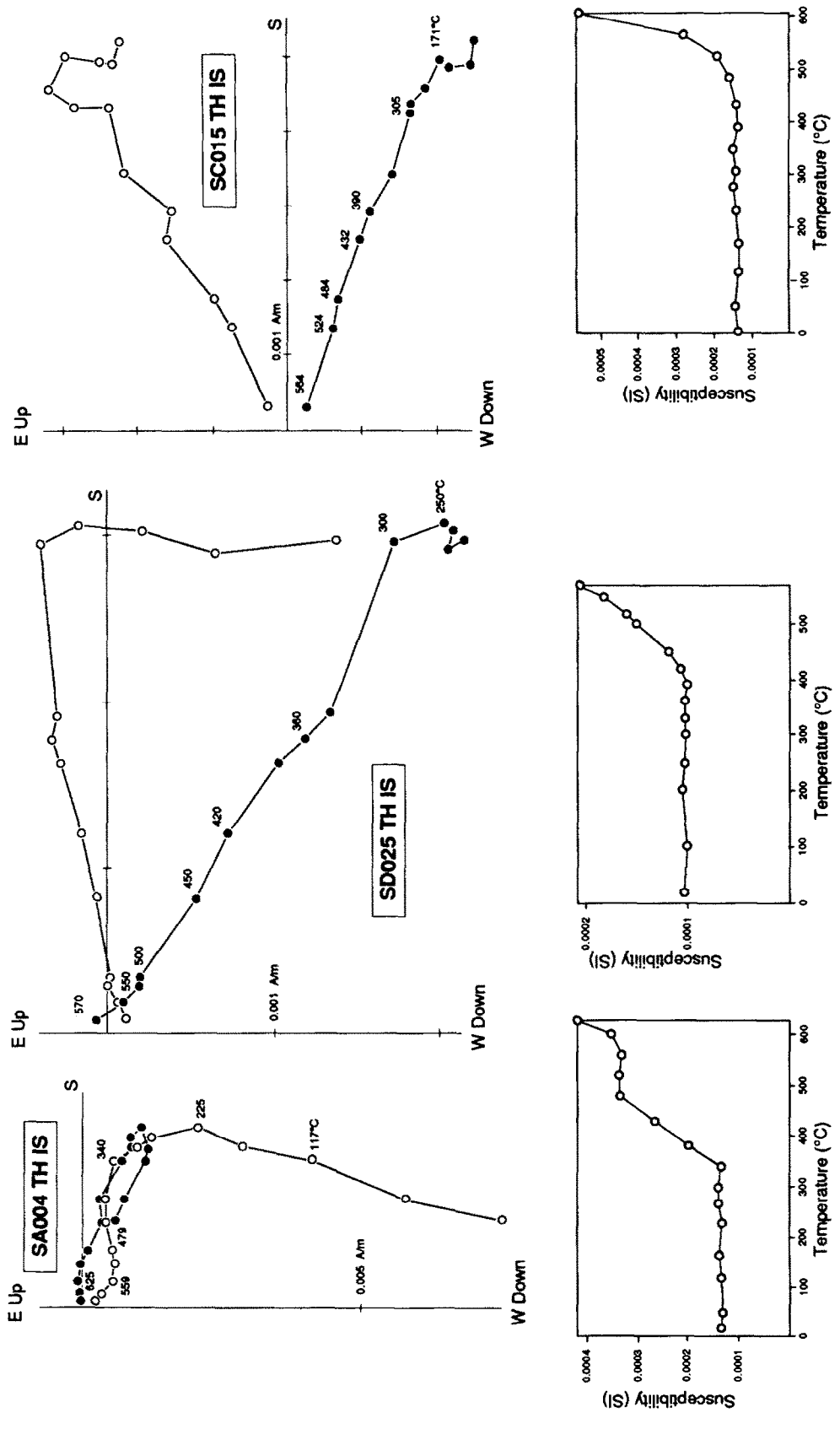


Fig. 10. Orthogonal projections of demagnetization curves from the Scania area. Same legend as Fig. 4.

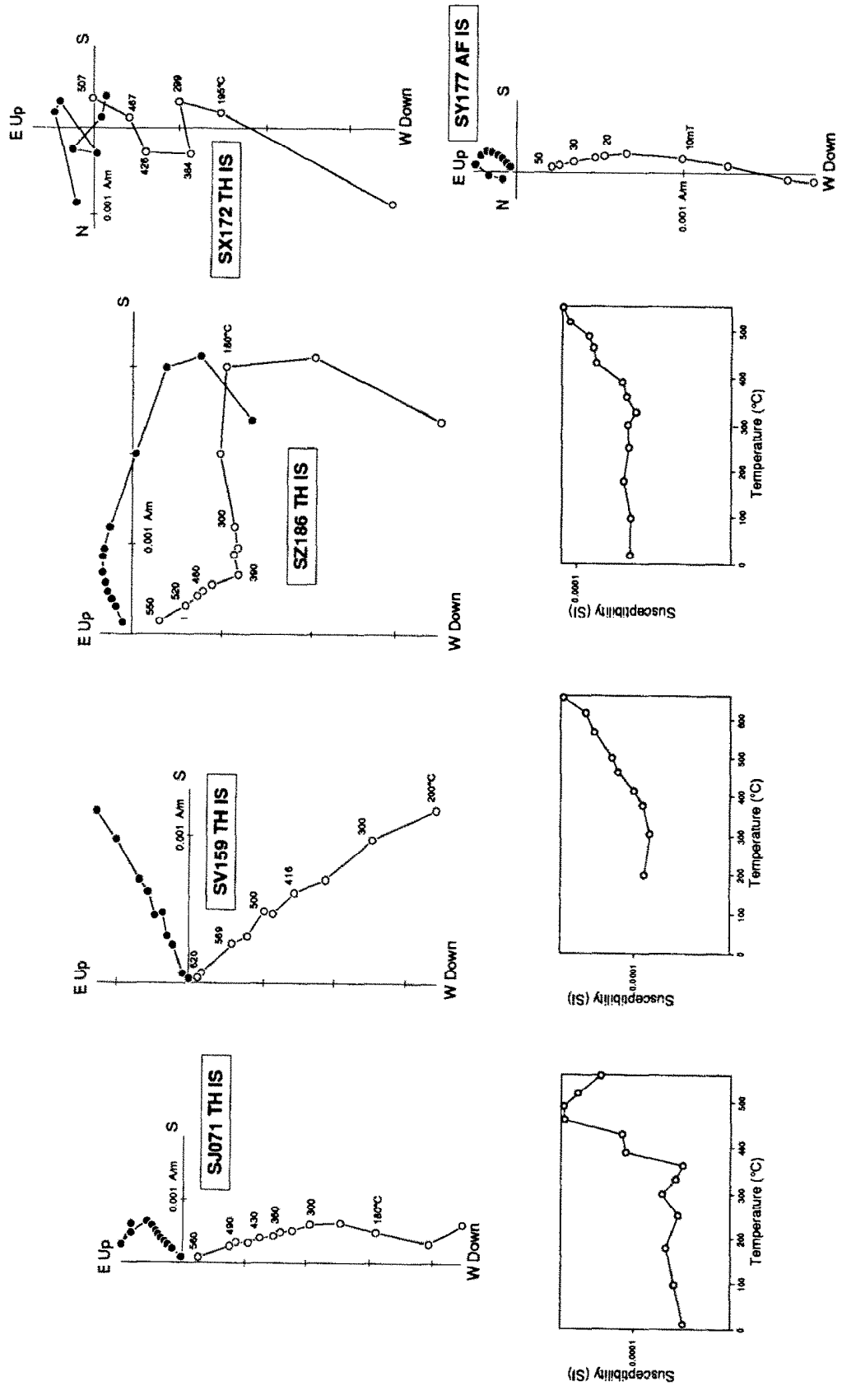


Fig. 11. Orthogonal projections of demagnetization curves from the other locations in Sweden. Same legend as Fig. 4.

TABLE 3

Palaeomagnetic results from Sweden

Sites	$Decl._{IS}$	$Incl._{IS}$	n	k	α_{95}	$Decl._{TC}$	$Incl._{TC}$	Observations
Scania								
SA	198	−1	9	90	6	198	−3	80–550 °C
SA	174	+56	7	67	7	169	+53	550–600 °C
SB	191	−21	6	350	4			dyke
SC	204	−22	11	192	3	206	−24	contact SB
SD	210	−13	8	467	3	210	−15	
Mean IS	202	−19	3	60	16			SB-SC-SD
Öland								
SE	121	+63	2	–	–	117	+61	
SF	138	+63	2	–	–	133	+62	
SG	143	+68	4	68	11	139	+67	
SH	141	+69	4	424	5	139	+67	
SI	137	+64	3	73	14	133	+65	
SJ	146	+69	4	634	4	145	+68	
SK	136	+66	3	1344	3	132	+65	
SL	155	+72	3	98	12	155	+72	
SM	140	+70	4	2339	2	140	+70	
SN	137	+70	4	756	3	137	+70	
SO	124	+67	4	673	4	116	+65	
Mean IS	137	+68	11	299	3			
Mean TC			11	212	3	134	+67	
Öster-götland								
SQ	126	+65	4	183	7	126	+65	
SR	128	+68	4	137	8	126	+66	
SS	137	+65	3	93	13	128	+66	
ST	118	+70	4	80	10	107	+69	
Mean IS	128	+67	4	436	5			
Mean TC			4	380	5	122	+67	
Dalarna								
SU	93	+70	3	333	7	85	+56	
SV	157	+48	5	647	3	142	+66	
SW	102	+45	4	282	5	100	+77	
Mean IS	121	+58	3	12	38			
Mean TC			3	26	25	107	+68	
Närke								
SX	117	+68	3	41	20			
SY	107	+70	4	46	14			
Mean	112	+69	2	–	–			
Väster-götland								
SZ	199	−15	4	91	10			250–430 °C
SZ	145	+54	4	69	11			430–550 °C
S1	195	−4	4	50	13			200–380 °C
S1	173	+50	4	47	14			380–550 °C
S2	196	−3	4	78	10			250–430 °C
S2	159	+59	4	116	9			430–550 °C
Mean Low T°	197	−7	3	135	11			
Mean High T°	159	+55	3	73	14			

Notes: see Table 1 for explanations of abbreviations.

the Mjøsa Limestone cluster around a northeast and up direction after tilt correction (decl. = 53°, incl. = −40°, α_{95} = 25°, 4 sites), with a significative increase of k from 6 to 14, although site-mean directions from sites ND and NE, sampled in the two limbs of a gentle syncline, bypass each other during unfolding. The acquisition of that component is therefore probably anterior to or at least synchronous with the Caledonian folding. Site NF, from the "*Orthoceras* limestone", displays a slightly different result; as it is a single site, it will not be considered in the final interpretation.

South Sweden

Most areas where the lower Palaeozoic cover is exposed in south Sweden, were sampled (Fig. 9). All sites belong to the Lower Ordovician "*Orthoceras* limestone". NRM measurements from all samples are very consistent: intensities are grouped between 10^{-2} and 10^{-3} A/m and initial susceptibility around 10^{-4} SI. Most directions are very steep and down, although some southwest and shallow directions are present in Scania, where Permian doleritic dykes are numerous and CAI values high (4–5).

Thermal demagnetizations from the Scania samples (Fig. 10) confirm the important role of the Permian overprint in that area. The three samples shown in Fig. 10 belong to sites sampled at different distances from the dykes (SA at about 100 m, SD 10 m and SC within 1 m of the contact of dyke SB). Their magnetic behaviours appear to be clearly related to the proximity of the intrusion: sample SC015 has only one component of magnetization, parallel to the dyke magnetization (site-mean results are given in Table 3); it has no soft component and its susceptibility remains unchanged up to 500 °C during the heating experiments. On the contrary, sample SA004 reveals three components of magnetization: a large present-day field component between 0 and 350 °C, an intermediate component comparable to the Permian overprint between 350 °C and 550 °C, and a high-temperature component above 550 °C, representing less than 10% of the initial NRM. Susceptibility is quite unstable, with abrupt changes occurring as early as at 380 °C and later at 600 °C. Sample SD025 has an intermediate behaviour: the present-day field component is present but much less pronounced, the Permian overprint dominates the NRM, and if it was not possible to define a high-temperature component, the demagnetization trajectory clearly overshoots the origin and the last step at 570 °C tends to be parallel with site the SA high-temperature component. Susceptibility starts to increase at about 450 °C. These observations are very similar to the Oslo Region results described above. The Permian overprint stabilizes the magnetization of the limestone, converting viscous components into stable remanent magnetizations. The conversion mechanism produces no change in susceptibility and therefore does not require any enrichment in the iron content of the rock. As the magnetic mineralogy is affected (vanishing of the viscous magnetic grains), we propose that the overprinting mechanism is based on chemical transformations of the minerals present in the rock, triggered by the heating due to the volcanism.

All other localities reveal the same high-temperature components, steeply inclined downwards, in the southeast quadrant (Fig. 11). Unblocking temperatures are around 550 °C for grey samples (SJ071, Fig. 11) or above 600 °C for red samples (SV159, Fig. 11). Susceptibilities start to increase usually at about 400 °C, corresponding to the less remagnetized samples in Scania, and Permian overprinting is absent, except for Västergötland samples (SZ186, Fig. 11), where Permian dolerite sills are present on top of the lower Palaeozoic cover. Some pyrite-rich samples were unstable upon thermal demagnetization (SX172, Fig. 11). Af demagnetization was then used (SY177, Fig. 11) and reveals exactly the same component between 20 and 50 mT. Site-mean results combining thermal and af experiments are listed in Table 3.

As the rocks are usually flat-lying, fold tests are not possible on the whole collection. However, sites from Dalarna were sampled in a ring described as a Silurian impact structure, and beds dip outwards from the ring. The 3 site-mean directions converge when applying the tilt-correction (k increases from 12 to 26, Table 3) and it

appears therefore that the magnetization is older than the Siljan astrobleme.

A comparison of our results with those obtained by Claesson (1977) on the same rock units shows the following:

(1) In undisturbed areas, they are very similar (Fig. 12) even if our mean directions are much better grouped. The use of a larger number of sampling sites by Claesson in Öland led to identical results, but at the price of a much more extensive field work. In this case, the use of modern palaeomagnetic techniques allows a lower error to be obtained with less samples.

(2) In areas with Permian intrusive, the blanket demagnetization technique is not able to isolate the components, as illustrated by the Scania example (Fig. 12), where Claesson's results are spread along the great-circle joining the two components we have identified.

Discussion

All magnetizations determined in this study and the corresponding pole positions are summarized in Table 4 and Fig. 13. Pole positions determined from overprint components in the Oslo Region are quite consistent with the expected Early Permian pole (lat. 49° S, long. 336° E, Bylund, 1986). However, overprints obtained in Swedish samples are slightly offset from those of the Oslo Region. This may correspond to a slightly different acquisition age (Late Carboniferous mean pole for Scandinavia at 38° S, 346° E, Bylund, 1986); another explanation can be an incomplete separation from the high-temperature components. In a similar way, the Swedish high-temperature components from undisturbed areas give very consistent pole positions, while those from Scania and Väster-götland are slightly offset

TABLE 4

Palaeomagnetic poles from this study

Area	S lat.	S long.	Decl.	Incl.	α_{95}	Plat.	Plong.	d_p	d_m	Polarity	Palaeolat.
Oslo Region											
Oslo–Asker	59.9N	10.5E	206	−45	3	52S	331E	2	4	R	27N
Slemmestad	59.8N	10.4E	213	−44	3	49S	322E	2	4	R	26N
Ringerike	60.1N	10.3E	192	−44	4	55S	351E	3	5	R	26N
Vikersund	59.9N	10.0E	198	−44	10	54S	342E	8	12	R	26N
Mjøsa	60.8N	10.5E	53	−40	25	04S	143E	18	30	N	23S
South Sweden											
Scania	55.6N	14.1E	202	−19	16	41S	345E	9	17	R	10N
			169	+53	7	00N	023E	7	10	R	34S
Öland	57.0N	17.0E	134	+67	3	23N	047E	4	5	R	50S
Öster-götland	58.4N	14.7E	122	+67	5	28N	053E	7	8	R	50S
Dalarna	61.0N	14.5E	107	+68	25	36N	063E	35	42	R	51S
Närke	59.1N	15.2E	112	+69	–	34N	058E	–	–	R	52S
Väster-götland	58.6N	13.4E	197	−07	11	33S	353E	6	11	R	04N
			159	+55	14	06N	030E	14	20	R	36S
Mean poles											
Overprint Oslo Region						53S	336E	$N=4$	$A_{95}=9$		
Overprint Sweden						37S	349E	$N=2$			
High T° Sweden						30N	055E	$N=4$	$A_{95}=9$		

Notes: S lat., S long. = site latitude and longitude; Decl., Incl. = declination, inclination of mean palaeomagnetic direction; α_{95} = confidence interval; P lat., P long. = palaeomagnetic pole latitude and longitude; d_p, d_m = oval of confidence about pole position; R, N = reverse, normal polarity.

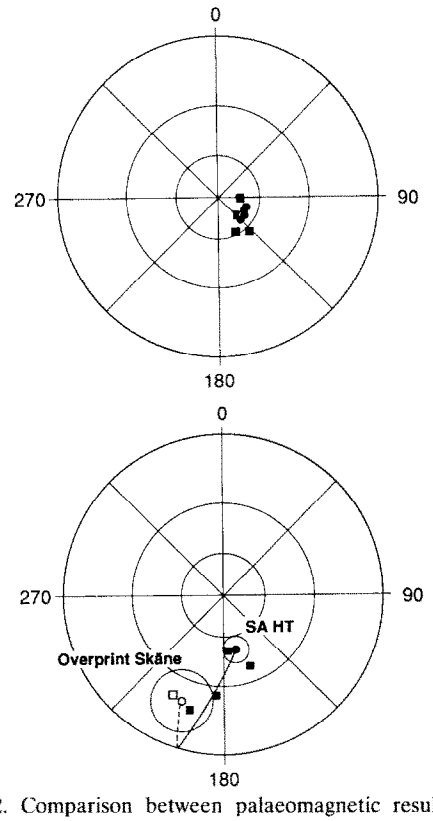


Fig. 12. Comparison between palaeomagnetic results from Claesson's study (1977, squares) and this study (dots) obtained in southern Sweden. Top: results from areas without Permian volcanic activity. Bottom: results from Scania.

towards the overprint poles. The same alternative can be advanced to interpret such results (age versus separation of components). In order to obtain the more accurate palaeomagnetic pole, we have kept for final calculation only the 4 poles from Öland, Närke, Dalarna and Öster-götland (pole position at 30 ° N, 55 ° E, $A_{95} = 9°$). The age of this pole is constrained by the age of the host rock (Lower Ordovician "*Orthoceras* limestone") and the age of emplacement of the Siljan asteroid. For the same rocks, Torsvik and Trench (1990) obtained positive foldtests from Claesson pilot samples in Öster-götland and Jämtland, confirming the acquisition age before the Caledonian folding. Also they discovered a few normal samples in Väster-götland with high-temperature components anti-parallel to our systematically reverse magnetizations. We conclude therefore that this pole corresponds probably to the primary magnetization of the "*Orthoceras* limestone" and we assign to it an Early Ordovician (~ 480 Ma) age.

In the discussion of her results, Claesson (1978) questioned their reliability arguing that she was not successful in isolating components. From our

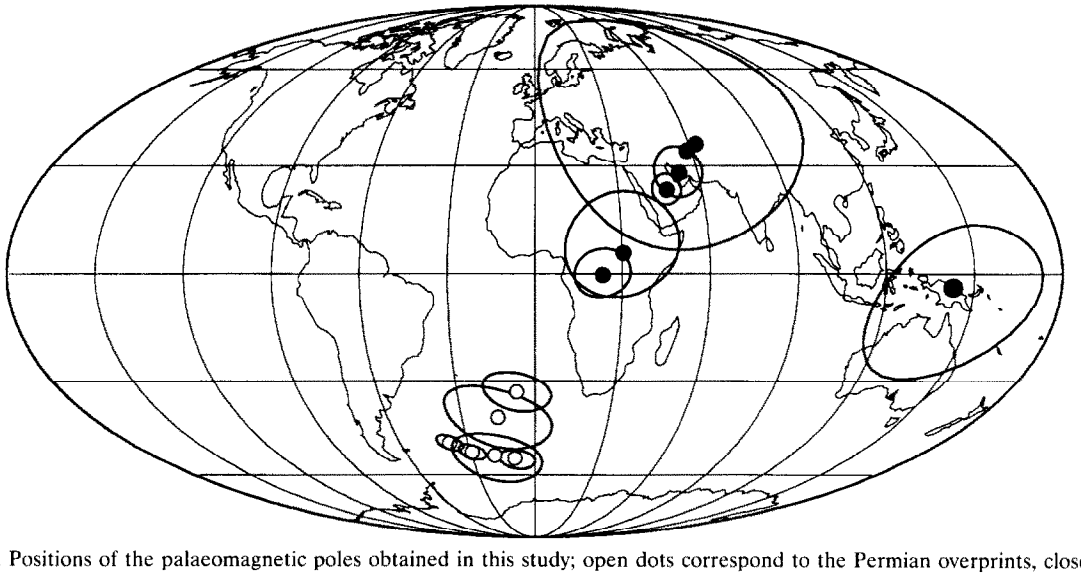


Fig. 13. Positions of the palaeomagnetic poles obtained in this study; open dots correspond to the Permian overprints, closed dots to the high temperature components. All poles are south poles except the Mjøsa Limestone pole which is north pole.

analysis, it seems that the high-temperature components in all four selected areas were very well defined by the demagnetization experiments. Moreover, thermal and af results concord perfectly. The only areas where such problems are real, are where the Permian overprint was detected (Scania and Väster-götland). We are confident that our selected results are fully reliable. In standard palaeomagnetic classification, they will probably rank at the highest level.

The pole position obtained from the Upper Caradoc Mjøsa Limestone in the north of the Oslo Region is quite different from the "*Orthoceras* limestone" pole. It is based on a small number of sites, and the dispersion is larger. Its isolation from secondary components by laboratory experiments is quite difficult. It cannot therefore be considered as having the same reliability level. However, tilt correction seems to improve the grouping of the site-mean directions, which allows to constrain its acquisition age in the interval Late Ordovician–Late Silurian. Furthermore, this pole does not resemble previously published poles of younger age from the Baltic Shield (Bylund, 1986), although similar poles were obtained in the British Isles with a Siluro–Devonian age (Torsvik et al., 1990). We therefore consider it as a preliminary but valuable estimation of the palaeo-field in Norway for the period of Late Ordovician to Late Silurian (~ 400–440 Ma), though further work is needed, at least, to reduce the confidence interval and precise its age.

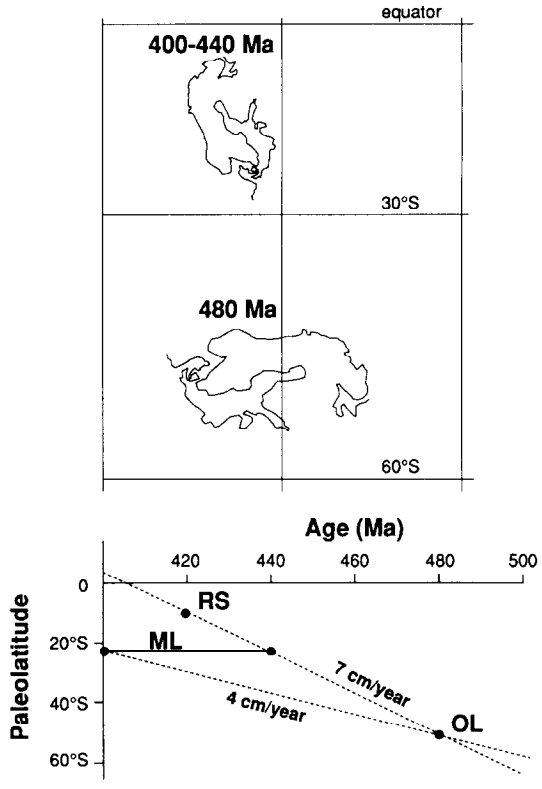


Fig. 14. Palaeogeographic reconstruction of Baltica during the Lower Palaeozoic, and corresponding palaeolatitude evolution with time (*OL* = "*Orthoceras* limestone", *ML* = Mjøsa Limestone; *RS* = Ringerike Sandstone). According to age uncertainty for the ML pole, drift rate can be estimated in the range 4 to 7 cm/yr; the highest rate, corresponding to an age of 440 Ma for ML, is perfectly consistent with the RS pole; on the contrary, a 400 Ma age for ML would imply more complex palaeolatitude evolution to fit the RS pole.

Conclusions

The palaeomagnetic results obtained in this study partially resolve the previous lack of knowledge about the palaeogeographic position of Baltica in the lower Palaeozoic. From the Lower Ordovician "*Orthoceras* limestone", we obtain a palaeolatitude of about 50 ° S at 480 Ma, while the Upper Ordovician Mjøsa Limestone indicates a palaeolatitude of 23 ° S at 400–440 Ma (Fig. 14). It seems therefore that the Baltic Shield has undergone a quite rapid (4 to 7 cm/yr) northward drift during this period of time, from a temperate latitude to the tropics. New Silurian results from the Ringerike Sandstone (Douglass, 1988) are perfectly consistent with the highest drift rate (7 cm/yr, corresponding to an age of 440 Ma for the Mjøsa Limestone pole) and suggest it might have extended at least up to the Silurian (420 Ma) (see Fig. 14). This conclusion agrees very well with the reconstruction proposed by Webby (1984) on the basis of the development of Late Caradoc reefs in Baltoscandia, or with the scenario based on the sedimentological study of the Ordovician limestones described in the first part of this paper. It seems therefore that all palaeogeographic methods are in agreement with regard to the position of Baltica during the Ordovician.

The demonstrated Ordovician northward drift was accompanied by a large rotation of Baltica, as illustrated by Fig. 14. The Euler pole being unknown, the rotation angle cannot be determined. However, it produces a change in declination from about 120° (reverse polarity) to about 50° (normal polarity), that represents an anticlockwise shift of 110° between Early Ordovician and Late Ordovician–Late Silurian.

The fairly high latitude of Baltica in the Early Ordovician (50° S) implies some changes in previous palaeogeographic reconstructions based upon palaeomagnetic data (Perroud et al., 1984; Briden et al., 1988; Torsvik et al., 1990). The question of the existence and width of the Medio-European (or Rheic) Ocean in the Ordovician, for example, needs to be questioned. However, this discussion is beyond the scope of this paper. It can simply be noticed that the width of this ocean in the Lower Ordovician was probably much narrower than previously thought. However, in the Upper Ordovician, a large distance might have separated Baltica and southern Europe, as represented by the Armorican terranes, unless the ages of the palaeomagnetic poles are incorrect. Finally, it has to be emphasized that these new palaeomagnetically determined palaeogeographical positions fit the biostratigraphical data well, and this should encourage the multi-disciplinary approach to the problem of reconstructing the successive geographies of the Earth in the geological past.

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Appendix—sampling sites

(1) NORWAY

Oslo–Asker–Slemmestad area
Stratigraphy, see Bockelie (1982).

The entire Ordovician succession has been sampled from the base to the top. All sampling sites are located either on the shore of the Oslofjord or in road sections along Route 165 between Holmen and Slemmestad.

OA–OF—Fornebu profile
OG–OH—Road 165, cross-roads at Vettre
OI—Björkas
OJ–OM—Vollen harbour (OJ north side, OK–OM south side)
ON–OP—Holmen Road 165 (east side of the road, to the north of Holmen Fjordhotell)
OQ—Ashgill channel fill on Nesöya
OR–OT—Road 165, 100 m north of cross-road to Heggedal (east side of the road)
OU–OW—Road 165, (east side) south of the cross-road to Heggedal
OX–OZ—idem (west side)
01–04—Road 165, Rortunet profile

Vikersund (Modum) area
Stratigraphy, see Wandas (1982)
RH—Øren farm
RI—south of Hovland
RJ–RL—near Vikersund ski-jump

Ringerike area
Stratigraphy, see Hanken and Owen (1982)
RA–RC—Bønsnes, on the shore of Tyrifjord
RD–RG—road section on route E68 to the south of "Botilrudveien bus stop".

Mjøsa district
Stratigraphy, see Opalinski and Harland (1981), Spjeldnæs (1982)
NA–NC—Toten area; Boverbru Kalkverk: quarry east of Boverbru railway station
ND–NE—Nes–Hamar area; Furuberget quarry
NF—Nes–Hamar area; road section along road E6 northwest of Hamar

(2) SWEDEN

Scania
Stratigraphy, see Bergström (1982)
Studied area, to the NNE of Smedstorp

SA–SC—Killerod quarry
SD—Flagabro farm (cf. Claesson's site Sk 19)
Öland
Stratigraphy, see Jaanusson (1960)
SE–SF—Segerstad, along the shore, north of the light-house
SG—Skarlov, stream (Storbrobäcken) section, E of the road from Skarlov to Segerstad
SH—Gillberga, quarry near the coast, South of Grytehamn (cf. Claesson's sites Ol-61-72)
SI—Gunnarstorp, abandoned quarry (50 m from Claesson's sites Ol-15-19)
SJ—Djupvik, abandoned quarry along the coast line, about 500 m south of Djupvik camping site
SK—Aleklinta, abandoned quarry to the SSE of the village (near Claesson'sites Ol-1-9)
SL–SN—Hornsudde, northern part of the cliffs (cf. Claesson's sites Ol-21-31)
SO—Binnerbäck, abandoned quarry to the NW of Binnerbäck, west of the main road
Öster-götland
SQ—Barstad, working quarry
SR—Borghamn, large abandoned quarry (cf. Claesson's sites Og-5-17)
SS–ST—Lyckhem, abandoned quarry
Dalarna
SU—Beriol (? Berihol), abandoned quarry in the forest (Claesson's site D6)
SV—Leskusäng (? Leskusänget), abandoned quarry in the forest (Claesson's sites D3-4)
SW—Sjurberg, railway cutting west of Rättvik (described in Tjernvik, 1956 pp. 165–167)
Närke
SX–SY—Yxhult, Hallabrotet quarry (cf. Claesson's sites Na 1–3)
Väster-götland, Kinnekulle
SZ–S1—abandoned quarries along road, east flank of Kinnekulle
S2—Hällekis abandoned quarry, lower part (cf. Claesson's sites Vg-15-30)

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